

SARCOMERE

DEFINITION

- **Sarcomere is defined as** the structural and functional unit of skeletal muscle.
- **Basic contractile unit** of the muscle.

EXTENT

- Sarcomere extends **between two Z-lines** of myofibril (*each myofibril contains many sarcomeres*).
- **In relaxed state:** Average length is **2 – 3 μm** .

COMPONENTS

- Each myofibril consists of an alternate:
 - **Dark 'A' band** (Anisotropic band)
 - **Light 'I' band** (Isotropic band)
- In the middle of the 'A' band, there is a lighter area called the **H-zone**.
- In the middle of the H-zone lies the middle part of the myosin filament called the **M-line** (*contains myosin-binding proteins*).

ELECTRON MICROSCOPIC STUDY OF SARCOMERE

- Sarcomere consists of many thread-like structures called **myofilaments**:
 - a. **Actin filaments** (Thin filaments)
 - b. **Myosin filaments** (Thick filaments)

1. ACTIN FILAMENTS

- **Thin filaments:**

- **Diameter:** 20 Å
- **Length:** 1 µm
- Extend from either side of the **Z-lines**, run across the **'I' band**, and enter into the **'A' band** up to the **'H' zone**.

Actin Molecule Components:

- a. **F-Actin (Fibrous Actin):** Double-stranded helix that possesses active sites.
- b. **G-Actin (Globular Actin):** Polymer of small globular proteins.
- c. **Tropomyosin:** Double-stranded protein wrapping around F-actin.
- d. **Troponin:** Complex of three subunits:
 - **Troponin T (T):** Binds to tropomyosin
 - **Troponin C (C):** Binds to calcium ions (Ca^{2+})
 - **Troponin I (I):** Inhibits actin–myosin interaction

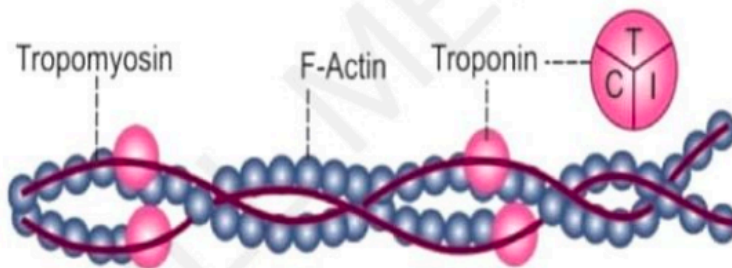


FIGURE 29.8: Part of actin filament. Troponin has three subunits, T, C and I.

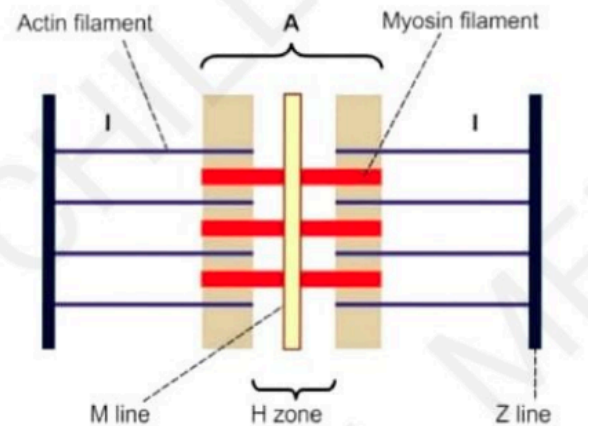


FIGURE 29.4: Sarcomere. A = A band, I = I band.

2. MYOSIN FILAMENTS

- **Thick filaments:**
 - **Diameter:** 115 Å
 - **Length:** 1.5 µm
- Situated in the **'A' band**.

Myosin Molecules (~200 per filament):

- Belongs to **Myosin II class** present in sarcomere.
- Composed of:

- a. **Tail Portion:** Made up of two heavy chains (H-chains) coiled together.
- b. **Head Portion:** Made up of the end portion of two heavy chains (globular head portion) plus four light chains.
 - Contains an **Actin-binding site**.
 - Contains an **ATP-binding site** (ATPase activity).

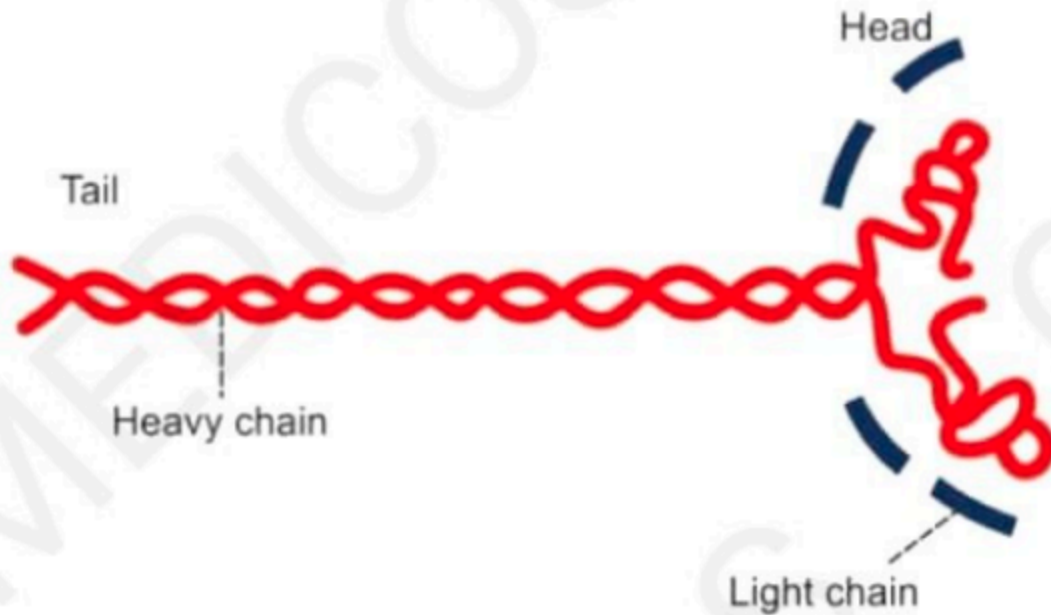


FIGURE 29.7: Myosin molecule formed by two heavy chains and four light chains of polypeptides

CROSS-BRIDGES & SLIDING MECHANISM

- **Myosin heads attach themselves** to the actin filaments forming cross-bridges.
- **Sliding Mechanism / Ratchet Mechanism:**
 - The head tilts and pulls the actin filament towards the center of the sarcomere during muscle contraction.

Resting Muscle

Z-lines at normal distance

H-zone and I-bands clearly
visible

A-band length constant

Myosin pulls Actin (Sliding)

Contracted Muscle

Z-lines pulled closer
together

H-zone disappears /
narrows

I-band width reduces

A-band length remains
unchanged

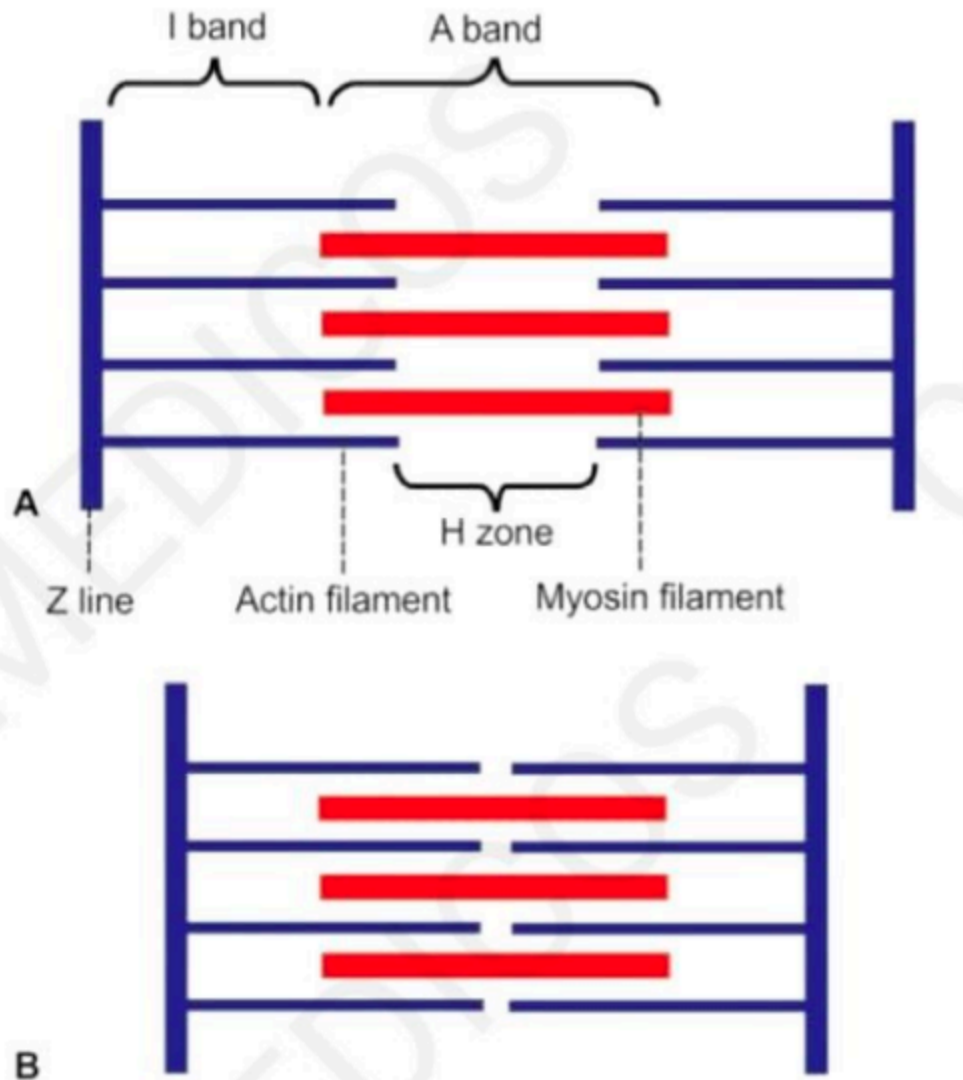


FIGURE 29.5: Sarcomere in resting muscle **A**. Contracted muscle; **B**. During contraction; Z lines come close, H zone and I band are reduced and no change in A band.

Changes in Banding During Contraction:

- a. **Z-lines:** Come closer together.
- b. **H-zone:** Reduces or completely disappears.
- c. **I-band:** Length reduces / narrows.
- d. **A-band:** Length remains unchanged.